Technical Document #259: Retrieval Augmented Generation - Comprehensive Overvi

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Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Query decomposition breaks complex queries into simpler sub-queries.
- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.
- Chunking splits documents into smaller segments for embedding.